COMPREHENSIVE BUSINESS REPORT

Olist E-Commerce Business Insights

Delivery Performance Analysis, Customer Satisfaction Metrics & RFM Customer Segmentation for Brazilian Marketplace

Data Overview

Foundational context: dataset origins, timeframe, and technological stack.

Dataset & Source

Brazilian E-Commerce Public Dataset by Olist

kaggle.com/datasets/olistbr

Analysis Period

24 Months

Sep 2016 — Sep 2018

Technologies Stack

Python | Pandas | SciPy | Matplotlib | Tableau Public

Project Details

Access project assets and team organisational structure.

Repository & Dashboard

GitHub Repository →

github.com/antik1108/...

Tableau Dashboard →

public.tableau.com/...

Team G-8 · Section C

Faculty Mentor

Archit Raj

Project Team

The dedicated individuals responsible for data analysis and reporting.

Antik Mondal — Team Lead / GitHub

Aadit Vachher — Tableau Dashboards

Anand Mishra — ETL & Data Cleaning

Sanchit Garg — EDA & Insights

Viraj Chafale — Statistical Analysis

Vridhi Chaudhary — Report & Presentation

Contents

1 Executive Summary **2**

2 Sector & Business Context **2**

3 Problem Statement & Objectives **2**

4 Data Description **3**

5 Data Cleaning & ETL Pipeline **3**

6 KPI & Metric Framework **3**

7 Exploratory Data Analysis (EDA) **4**

8 Statistical Analysis **5**

8.1 Hypothesis Test — On-Time vs. Late Review Scores (Welch's T-Test) 5

8.2 Pearson Correlation — Delay Magnitude vs. Review Score 6

8.3 RFM Customer Segmentation 6

9 Tableau Dashboard Design **7**

10 Insights Summary **8**

11 Recommendations **8**

12 Impact Estimation **8**

13 Limitations **8**

14 Future Scope **9**

15 Conclusion **9**

Appendix **10**

Contribution Matrix **11**

1. Executive Summary

Olist's marketplace serves 94,990 customers via 3,095 sellers across 27 Brazilian states. With R$13.5M in GMV across 2016–2018, the platform is commercially strong — but a measurable delivery performance gap is silently destroying customer satisfaction and accelerating churn. This analysis quantifies the damage and prescribes the fix.

93.3%	4.12	R$13.5M	43.7%
On-Time Delivery	**Avg Review Score**	**Total GMV**	**Customers at Risk**
6,534 late orders	Late orders: 3.27	2016–2018	of churn or lost

Problem: A structurally concentrated 6.7% late-delivery rate is producing statistically irrefutable customer satisfaction damage ($t = 98.38$, $p < 0.001$). Late orders score **0.98 stars lower** on average. Meanwhile, 41,528 customers are at risk of permanent churn.

Approach: Eight raw Olist tables merged and cleaned in Python (98,207 rows, 36 columns). Statistical hypothesis testing and RFM segmentation performed. All outputs visualised in Tableau Public.

Key Findings:

- Late deliveries generate review scores of 3.27 vs. 4.25 for on-time orders — a gap proven non-random at $p < 0.001$.
- São Paulo alone generates 39% of GMV; top 3 states exceed 60%. Northern states face the longest delays.
- Pearson $r = 0.27$: every 5-day delay increase = approximately 0.4 stars in expected review score.
- Only 6,126 customers (6.4%) are Champions; 43.7% are At Risk, High Risk/Churn, or Lost.

Key Recommendations:

1. Negotiate SLA tiers with regional 3PL partners in Northern/Northeastern states.
2. Launch targeted win-back campaign for 29,502 At Risk and High Risk/Churn customers.
3. Introduce a seller accountability scorecard — surface on-time rate and review score monthly.

2. Sector & Business Context

Brazil is Latin America's largest e-commerce market (**USD 50B+ annually**). Between 2016–2018, marketplace aggregators like Olist enabled thousands of SMBs to sell nationally via a single contract — creating scale but also logistics complexity across 8.5 million km².

Industry Challenges

- Continental geography creates 3–5× delivery time differentials between Southeast and North.
- Public review systems make satisfaction scores a permanent competitive signal.
- High customer acquisition cost (CAC) makes repeat purchase rate the core profitability lever.
- Customers expect Amazon-standard delivery reliability as a baseline.

Decision-Makers & Business Value

This report serves Olist's **COO** (logistics SLA negotiations, carrier investment) and **Head of Seller Success** (seller accountability design). Delivery is chosen as the focal problem because it is the only first-physical-touchpoint Olist fully controls. A 1pp improvement in OTDR eliminates ≈980 unhappy customers per year and predictably lifts platform-wide review scores and GMV.

3. Problem Statement & Objectives

"Olist's marketplace generates R$13.5M GMV across 98,207 orders, yet a geographically concentrated 6.7% late-delivery rate is systematically depressing customer satisfaction. This project quantifies the magnitude, drivers, and customer-level churn risk of delivery underperformance, and prescribes targeted operational interventions."

In Scope

- 98,207 orders, Sep 2016–Sep 2018
- On-time rate, delivery time, delay analysis
- State-level geographic performance mapping
- Review score correlation with delivery metrics
- RFM segmentation of 94,990 customers
- Hypothesis testing and Pearson correlation

Out of Scope

- Cancelled / unavailable orders
- Payment method analysis
- Real-time or predictive forecasting
- Seller-level financial/margin analysis
- Marketing funnel dataset (not joined)

Success Criteria: Welch t-test result at α=0.05; Pearson r computed; 27 states mapped; 94,990 customers segmented; Tableau dashboard accessible; all Python in GitHub.

4. Data Description

Source: Brazilian E-Commerce Public Dataset by Olist — kaggle.com/datasets/olistbr/brazilian-ecommerce — CC BY-NC-SA 4.0. Real anonymised commercial data, 2016–2018.

Raw Tables (8 files)

Table	Rows
olist_orders_dataset	99,441
olist_order_items_dataset	112,650
olist_customers_dataset	99,441
olist_order_reviews_dataset	99,224
olist_products_dataset	32,951
olist_sellers_dataset	3,095
product_category_translation	71
olist_geolocation_dataset	1,000,163

Master Dataset (post-cleaning)

Property	Value
File	olist_master_dataset.csv
Rows	**98,207**
Columns	**36**
Period	Sep 2016–Sep 2018
Unique customers	94,990
Total GMV	R$13,494,401
On-time rate	93.3%
Avg review score	4.12 / 5.0

Key Columns:

Column	Type	Description
order_id	string	Unique hash per order (primary key)
order_purchase_timestamp	datetime	Purchase time; used for time-series
order_delivered_customer_date	datetime	Actual delivery; used in delay calc
order_estimated_delivery_date	datetime	SLA promise shown at checkout
customer_unique_id	string	Cross-order ID for RFM segmentation
customer_state	string	2-letter Brazilian state code
product_category_name_english	string	Translated from Portuguese
revenue / total_price	float	Sum of item prices per order (BRL)
review_score	float	Customer rating 1–5; 734 nulls excluded
delivery_time_days	float	(delivered − purchase).days — derived
delivery_delay_days	float	(delivered − estimated).days — derived
is_late	bool	True if delivery_delay_days > 0

Limitations: 42% of orders from São Paulo; data ends Sep 2018; multi-item orders collapsed to single row; geolocation not merged at order level; 734 null review scores excluded.

5. Data Cleaning & ETL Pipeline

Reference: `notebooks/02_cleaning.ipynb`

Cleaning Steps

1. Load & audit 8 raw tables (4,908 nulls; 261,831 geo duplicates)
2. Portuguese→English translation via LEFT JOIN (29 unmapped → 'Unknown')
3. Aggregate items per order (SUM price/freight; FIRST product by price)
4. Deduplicate reviews (most recent per order)
5. Master LEFT JOIN across all 5 tables
6. Filter: remove `canceled/unavailable` (1,234 rows)
7. Null fills: comments → 'No Comment'; category → 'Unknown'
8. Datetime conversion with `errors='coerce'`
9. Outlier review: extreme delivery days retained
10. Feature engineering (5 derived columns)

Feature Engineering

New Column	Formula & Purpose
delivery_time_days	(delivered − purchase).days — total wait
delivery_delay_days	(delivered − estimated).days — SLA gap
is_late	delay_days > 0 — boolean flag
order_year_month	YYYY-MM period — time-series
order_year	Calendar year — YoY grouping

Null Treatment

Column	N	Treatment
review_comment	∼48k	→ 'No Comment'
category (English)	∼610	→ 'Unknown'
order_approved_at	19	Left null
delivered_carrier	624	Left null
delivered_customer	1,737	Left null; excl. KPIs
review_score	734	Left null; excl. avg

6. KPI & Metric Framework

KPI 1 — On-Time Delivery Rate (OTDR)

$$\text{OTDR} = \frac{\text{is_late = False}}{\text{total non-null is_late}} \times 100$$

Value: 93.3% (6,534 late / 98,207 total)

Segmented by seller_state, customer_state, product category.

Why: Measures SLA promise compliance. Below 90% triggers systematic CSAT decline.

KPI 2 — Weighted Avg Review Score (WARS)

$$\text{WARS} = \frac{\sum(\text{score} \times \text{orders})}{\sum \text{orders}} \text{ per seller/category}$$

Value: 4.12 (On-time: 4.25 | Late: 3.27)

Correlated with delivery_delay_days via Pearson r.

Why: Review score is a leading commercial signal — lower scores suppress search ranking.

7. Exploratory Data Analysis (EDA)

Reference: `notebooks/03_eda.ipynb`

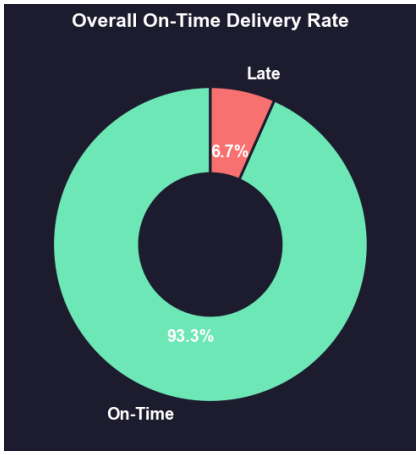


Figure 1. On-Time Delivery Rate

Insight: 93.3% on-time looks strong, but the 6.7% late cohort disproportionately drives negative reviews.

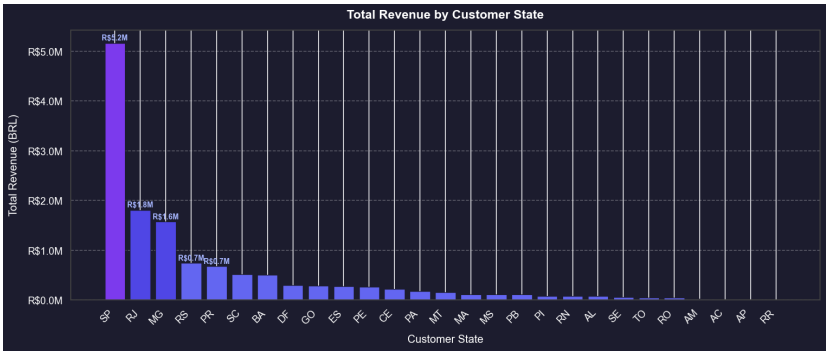


Figure 2. Total Revenue by Customer State (all 27 Brazilian states)

Insight: SP generates R$5.2M — 39% of GMV. Top 3 states (SP, RJ, MG) exceed 60%. Northern states generate near-zero revenue due to logistics infrastructure deficits.

Figure 3. Average Review Score by Product Category — Top 20 (green) and Bottom 10 (red)

Insight: `cds_dvds_musicals` and `books_general_interest` score $\geq$4.5. `security_and_services` falls below 3.0.

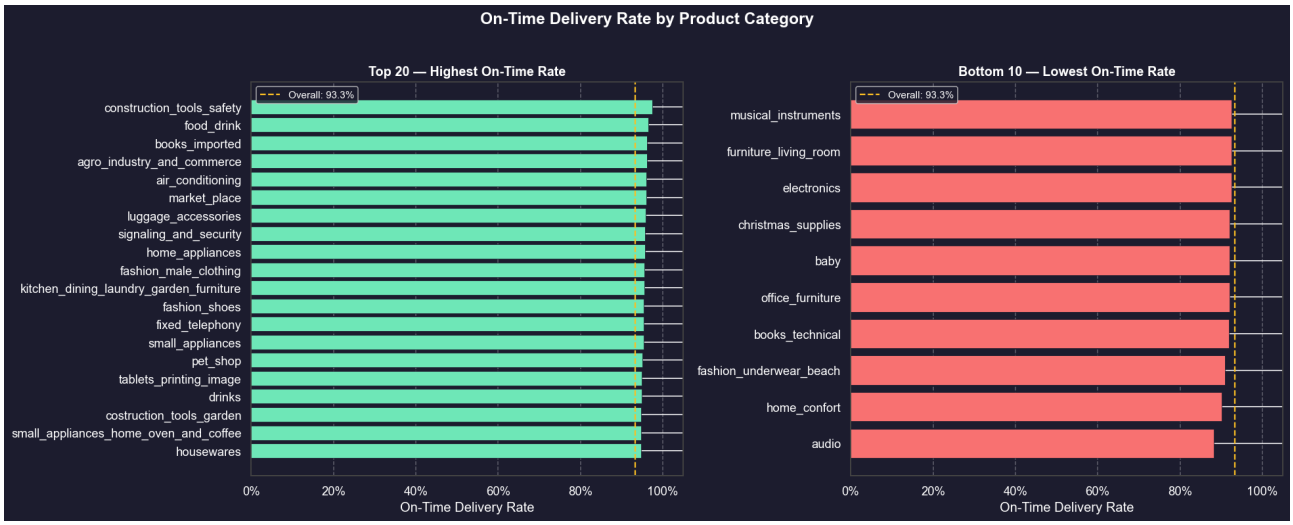


Figure 4. On-Time Delivery Rate by Category — Top 20 and Bottom 10; dashed = 93.3% benchmark

Insight: `construction_tools_safety`, `food_drink`, `books_imported` achieve near-100% OTDR. `musical_instruments`, `furniture_living_room`, and `audio` fall below benchmark.

8. Statistical Analysis

Reference: `notebooks/04_statistical_analysis.ipynb`

8.1 Hypothesis Test — On-Time vs. Late Review Scores (Welch's T-Test)

H_0: Mean review score of on-time orders = mean review score of late orders. **H_1:** On-time mean $>$ late mean. **Test:** Welch's two-sample t-test (one-tailed, $\alpha = 0.05$).

Metric	On-Time	Late
Sample N	91,673	6,534
Mean score	**4.25**	**3.27**
Std deviation	1.03	1.38
t-statistic	98.38	
p-value	< 0.001	
Decision	**REJECT H_0**	

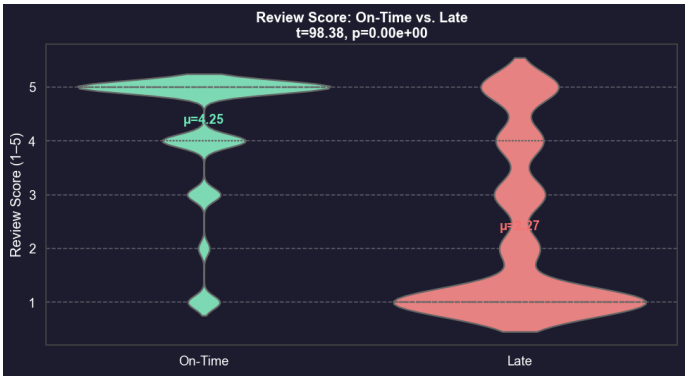


Figure 5. Review Score: On-Time vs. Late

Business Interpretation

$p < 0.001$ with t = 98.38 — the result is unambiguous. Late deliveries *systematically* depress customer satisfaction. The 0.98-star gap is not random. Every 1% reduction in late delivery rate produces a measurable, predictable uplift in platform ratings.

8.2 Pearson Correlation — Delay Magnitude vs. Review Score

Metric	Value
Pearson r	**-0.2673**
p-value	< 0.001
Sample N	≈96,470
Direction	Negative
Strength	Weak-moderate
Practical	-0.4 stars per +5 days

Interpretation: r = -0.27 is expected. Significance at $p < 0.001$ across 96k+ orders confirms the relationship is real and actionable.

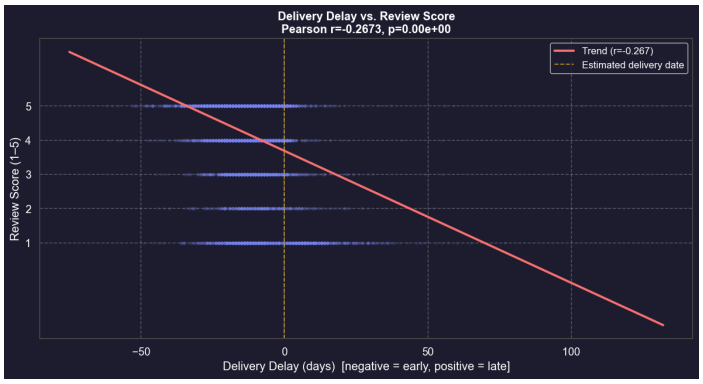


Figure 6. Delivery Delay vs. Review Score

8.3 RFM Customer Segmentation

Recency, Frequency, Monetary scored 1–4 via quartile ranking across 94,990 unique customers. Snapshot: 4 Sep 2018.

Segment	Customers	Avg Spend	Action
Loyal Customers	29,607	R$122.8	Cross-sell
At Risk	17,786	R$140.8	Re-engage
Lost	12,026	R$139.3	Reactivate
Potential Loyalists	11,858	R$137.5	Nurture
High Risk/Churn	11,716	R$145.0	Win-back
Champions	6,126	R$248.6	VIP rewards
Needs Attention	5,871	R$141.2	Survey

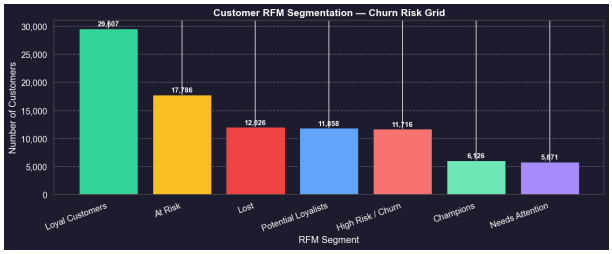


Figure 7. RFM Churn Risk Grid

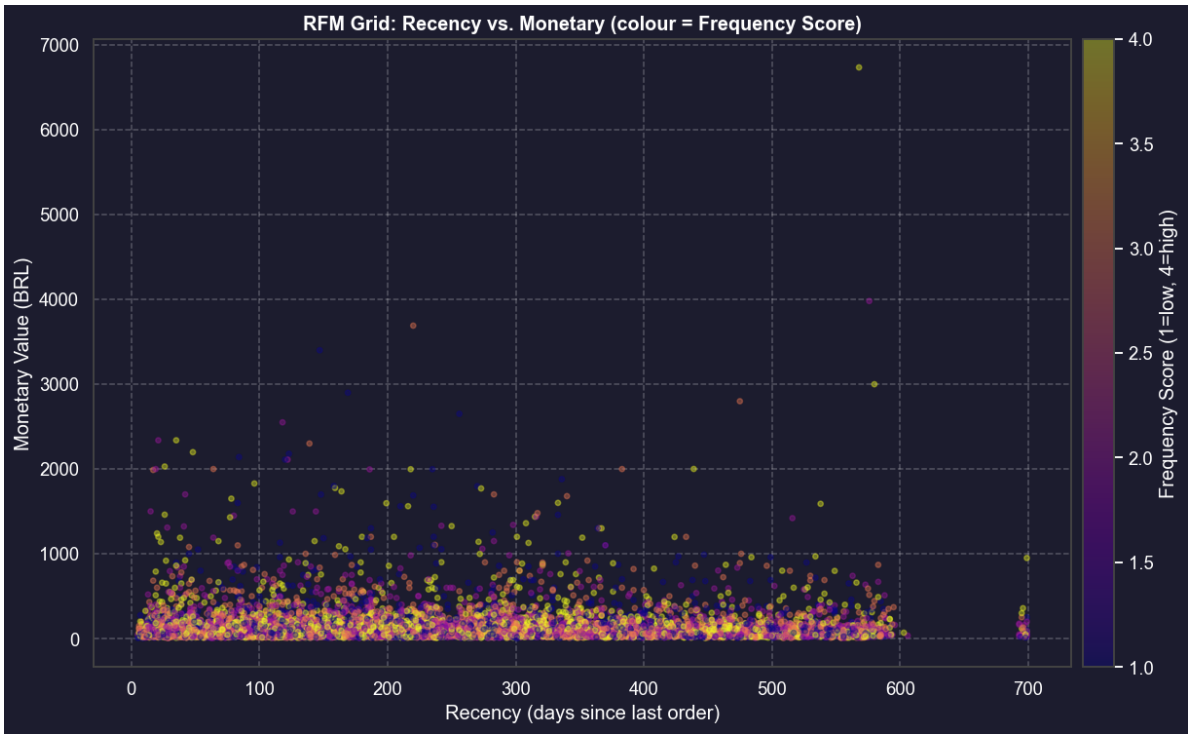


Figure 8. RFM Grid: Recency vs. Monetary (colour = Frequency Score)

Business Interpretation

Only 6,126 customers (6.4%) are Champions. A further 41,528 (43.7%) are At Risk, High Risk/Churn, or Lost. The At Risk cohort (R$140.8 avg spend, 283-day avg recency) is the highest-urgency win-back target.

9. Tableau Dashboard Design

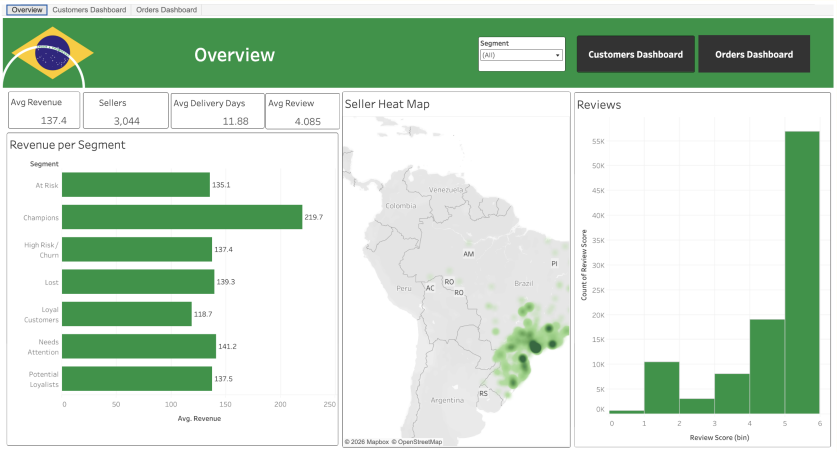
URL: public.tableau.com/app/profile/aadit.vachher/viz/brazile-commerceanalysis

Dashboard Objective: Answer: (1) *"How is the platform performing?"* — for the COO; (2) *"Which states, categories, and sellers are underperforming?"* — for Operations.

View	Charts Included	Audience & Purpose
Executive Overview	On-time donut, revenue by state bar, monthly GMV trend, review heatmap	COO, Commercial Director — platform KPI scorecard
Customer Segments	RFM bar chart, recency vs. monetary scatter, segment filter panel	Head of Marketing, CRM — churn risk management
Orders & Delivery	On-time rate by category (top/bottom 20), review by category, state delay map	Logistics Manager, Seller Success Team — operational drill-down

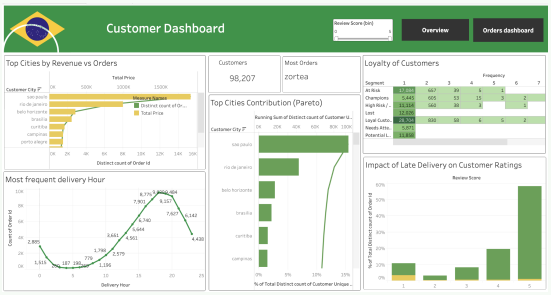
Dashboard Screenshots

View 1: Executive Overview COO / Commercial Director — Platform KPI Scorecard



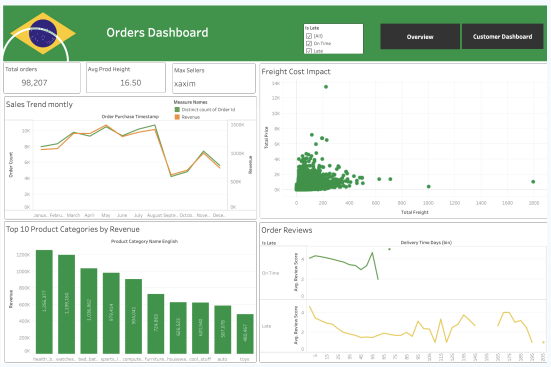
View 2: Customer Segments

Head of Marketing / CRM



View 3: Orders & Delivery

Logistics Manager / Seller Success



10. Insights Summary

1. **Late delivery is a predictable satisfaction destroyer.** 6,534 late orders average 3.27 stars vs. 4.25 for on-time ($t = 98.38$, $p < 0.001$).
2. **São Paulo's 39% GMV share creates single-point revenue risk** requiring geographic diversification via logistics investment in the North and Northeast.
3. **Delay magnitude directly predicts review score** ($r = -0.2673$, $p < 0.001$).
4. **Books, music, and safety tools deliver near-perfect on-time rates**; furniture, audio, and musical instruments consistently underperform.
5. `security_and_services` **scores below 3.0** — seller qualification must be tightened.
6. **Only 6.4% of customers are Champions**; the platform has a structural loyalty depth problem.
7. **43.7% of the customer base is in churn territory** (At Risk + High Risk/Churn + Lost).
8. **The At Risk cohort (17,786 customers, R$140.8 avg spend) is the most urgent win-back target.**
9. **Potential Loyalists (11,858 customers, 118-day recency)** are the platform's most scalable growth lever.
10. **Revenue grew from R$6.1M (2017) to R$7.3M (2018)** — demonstrating market expansion potential.

11. Recommendations

#	Recommendation	Expected Impact	Insight Source	Feasibility
1	**Fix logistics tail in N/NE states** — negotiate SLA tiers with regional 3PLs; establish micro-fulfilment	3,000+ fewer negative reviews/year; OTDR > 95%	Geographic delay analysis + $r = -0.27$	Medium (3–6 months)
2	**Win-back campaign** — 3-tier: At Risk (10% discount), High Risk/Churn (20% + free shipping)	R$245,000+ recovered revenue at 10% reactivation	RFM segments: 29,502 targets	Low-medium (30 days)
3	**Seller accountability scorecard** — surface monthly OTDR and WARS vs. benchmarks	2,940 fewer late deliveries/year; 3pp OTDR gain	Pearson correlation; KPI framework	Medium (60 days)
4	**Recalibrate delivery estimates** — add 3–5 buffer days for furniture/audio/instruments	≈2,000 technical "late" converted to "on-time"	Category OTDR analysis	Low (1 sprint)

12. Impact Estimation

Action	Type	Est. Impact	Logic
Logistics SLA (N/NE)	Risk reduction	3,000+ reviews/yr	50% improvement in bottom-10 states × current late volume
Win-back campaign	Revenue uplift	R$245k+ per cycle	10% reactivation × 1,750 At Risk × R$140 avg spend
Seller scorecard	Efficiency	2,940 late orders/yr	3pp OTDR gain at current 98k order volume
Delivery date recalibration	CSAT gain	2,000 ratings prevented	Category OTDR gap × bulky-item order volume

Act Now: The At Risk cohort (17,786 customers, avg recency 283 days) will cross into permanent "Lost" status within 4 months. Win-back cost increases 3–5× once customers enter the Lost segment. The recovery window is open — but closing.

13. Limitations

Data Gaps

- Dataset ends Sep 2018 — post-2018 market changes not captured.
- 734 null review scores may be biased toward dissatisfied customers.
- No carrier-level data — cannot benchmark individual logistics partners.
- Geolocation not merged at order level.

What Cannot Be Concluded

- **Causality:** t-test and Pearson establish association, not causation.
- **Individual seller performance:** Analysis is platform/state level.
- **Review sentiment nuance:** 48k+ written comments not NLP-analysed.
- Pearson assumes linear relationship; true dynamic may be threshold-based.

14. Future Scope

Additional Analysis

- **NLP on reviews:** Sentiment analysis and LDA topic modelling.
- **Forecasting:** ARIMA/Prophet on monthly GMV.
- **Seller clustering:** Group sellers by delivery + review performance.
- **Survival analysis:** Model time-to-churn per RFM cohort.
- **Chi-squared:** Test independence between category and late-delivery.

New Data & Models

- **Carrier data:** Enable per-logistics-partner SLA benchmarking.
- **Post-2018 orders:** Capture pandemic surge and infrastructure impact.
- **Marketing funnel:** Join Olist Marketing Dataset.
- **Delivery risk model:** XGBoost classifier for late-risk prediction.
- **Real-time dashboard:** Tableau Online + live DB feed.

15. Conclusion

Olist's platform processes nearly 100,000 orders across Brazil with a 93.3% on-time delivery rate and R$13.5M in GMV — commercially strong, yet operationally fragile. This project proved with statistical certainty ($t = 98.38$, $p < 0.001$) that the 6.7% of customers receiving late deliveries rate the platform **0.98 stars lower**, driving a Pearson $r = -0.27$ relationship between delay magnitude and review score across 96,470 orders. Geographic analysis revealed that the Southeast (SP/RJ/MG) generates over 60% of GMV while Northern states suffer structurally longer delays. RFM segmentation identified 41,528 customers (43.7%) already in churn territory, contrasted against only 6,126 Champions. The path forward is clear: fix the logistics tail in underserved regions through targeted 3PL contracts, launch an immediate win-back campaign for the 29,502 At Risk and High Risk/Churn customers (recoverable R$245k+ at 10% reactivation), and introduce seller accountability infrastructure. These three actions, executed within 90 days, represent the highest-ROI path to improving both operational performance and long-term commercial growth for the Olist platform.

Appendix

A. Full Data Dictionary

Column	Type	Description & Cleaning Notes
order_id	string	Unique hash; primary key; no duplicates
customer_id	string	Hash linking to customer record
order_status	string	Filtered: removed canceled/unavailable
order_purchase_timestamp	datetime	Converted from string; time-series base
order_delivered_customer_date	datetime	1,737 nulls; used in delivery time calc
customer_unique_id	string	Cross-order identifier for RFM
customer_state	string	2-letter code (uppercase)
product_category_name_english	string	Translated; Unknown for unmapped
total_price / revenue	float	SUM of item prices (BRL)
review_score	float	1–5 rating; 734 nulls excluded
delivery_time_days	float	(delivered − purchase).days
delivery_delay_days	float	(delivered − estimated).days
is_late	bool	delay_days > 0

B. RFM Segmentation Logic (Python)

```
def rfm_segment(row):
r, f, m = row['R'], row['F'], row['M']
if r >= 4 and f >= 3 and m >= 3:  return 'Champions'
elif r >= 3 and f >= 2:      return 'Loyal Customers'
elif r >= 3 and f == 1:      return 'Potential Loyalists'
elif r == 2 and f >= 2:      return 'At Risk'
elif r <= 2 and f >= 3:      return 'High Risk / Churn'
elif r == 1:                 return 'Lost'
else:                        return 'Needs Attention'
```

C. Cleaning Log Summary

- Input: 99,441 raw orders across 8 tables.
- Removed: 1,234 canceled/unavailable orders → 98,207 final rows.
- Filled: ∼48,000 null review comments → 'No Comment'; ∼610 null categories → 'Unknown'.
- Engineered: 5 derived columns (delivery_time_days, delivery_delay_days, is_late, order_year_month, order_year).
- Output: 98,207 rows × 36 columns → `olist_master_dataset.csv`.

Contribution Matrix (Mandatory)

Claims must match evidence in GitHub Insights, PR history, and committed files.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Aadit Vachher	✓	—	—	✓	✓	—	✓
Anand Mishra	✓	✓	✓	✓	—	—	✓
Antik Mondal	✓	✓	—	✓	—	✓	✓
Sanchit Garg	✓	—	—	✓	✓	—	✓
Viraj Chafale	✓	—	✓	✓	—	—	✓
Vridhi Chaudhary	✓	—	—	✓	—	✓	✓

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artefacts.

Team Lead Signature: Antik Mondal **Date:** 28/04/2026